THE CERTIFIED BOUNDARY EDITION

AGI ALPHA FLYWHEEL

When Capability Creates Opportunity

Transformational Intelligence, Experience-Trained Specialists and the Expanding Frontier of Profitable Work

An executable answer to a covering-system question

$$h \geq \frac{17902560}{18258529} > 0.98$$

Exact proof · Independent checking · Honest economics

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TRANSFORMATIONAL INTELLIGENCE: LEARNING TO CHANGE THE GAME

FIND YOUR ALPHA. BUILD YOUR SUCCESSOR Ω.

27 October 2026 · Research-review edition v1

A question that now has a checkable answer

An unbounded search has become a finite, reusable proof. This study constructs an exact mathematical certificate that rules out a proposed family of low-heft congruence partitions for every interval length. A reader can check the result with ordinary Python, without a language model, a numerical optimizer, a wallet or a trusted prediction about future capability. The artifact is the useful outcome: a precise answer that subsequent researchers can inspect, challenge and build upon.

The problem is simple to state. Partition the integers from 1 to n into complete residue classes, with each class containing at least two integers of that interval. A class with modulus m contributes $1/m$ to the partition's **heft**. Gerry Myerson asks whether, with distinct moduli, such a partition can have heft below 0.98 (Myerson 2026; Myerson, Poon, and Simpson 2009). We establish the following computer-assisted theorem:

$$h = \sum_i \frac{1}{m_i} \geq J := \frac{17902560}{18258529} = 0.9805039606421744\ldots > 0.98.$$

The strongest certificate was discovered by the fresh conventional challenge and strengthened by exact normalization in independent review. The proof covers every positive integer n admitting a partition and even permits repeated moduli. It preserves complete residue classes, exact coverage and at least two hits per class. It does not apply to arbitrary selected arithmetic progressions, singleton classes or two-dimensional coverings. The inspected prior general lower bound is $\sqrt{8} - 2 \approx 0.828427$, from Emanuel's dissertation, Appendix C, Theorem C.1 (Emanuel 2011). The constant above is certified, not claimed optimal. A bounded literature audit did not locate this result; external mathematical review and a fuller priority assessment remain pending.

The enterprise conclusion is narrower than the scientific result. No customer bought the work; actual revenue and receipts are zero, and complete actual authoring cost is unknown. An explicit planning model for review and integration of this existing artifact is conditionally profitable with four compatible sponsors, but demand is unvalidated and financing is required. A competent conventional shared arrangement has the same economics. No measured Operational Alpha, superhuman specialist, profitable skill compounding or general discovery engine is established.

Abstract. We investigate whether a bounded, independently challenged discovery process can produce a new useful result rather than an architectural proposal or an attractive scenario. A diverse opportunity screen rejects generic novelty claims already addressed by established methods. A first exact-code construction lane fails to improve the strongest located comparators. A subsequent finite-cover investigation first constructs an exact degree-20 dual weight. A fresh conventional challenge independently discovers a simpler piecewise-linear weight; normalization by its exactly checked maximum gives the stronger final bound. Two separately implemented line-arrangement checkers and an analytic tail certify every allowed class, while cell integrals transfer the result to every finite interval length. Fresh tests examine additional classes, scope decisions and malformed or altered inputs. The earlier polynomial proof remains independently verified and reproducible. Failed numerical evidence is retained. Existing AGI ALPHA provenance and execution components are exercised locally without minting, live identity claims or financial activity. Complete conditional accounting separates operating surplus, full-project profit, funding needs, restricted deposits and service obligations. The result demonstrates a portable research capability and a concrete mathematical advance relative to inspected prior bounds. It does not demonstrate an exclusive algorithmic advantage or a commercially validated enterprise.

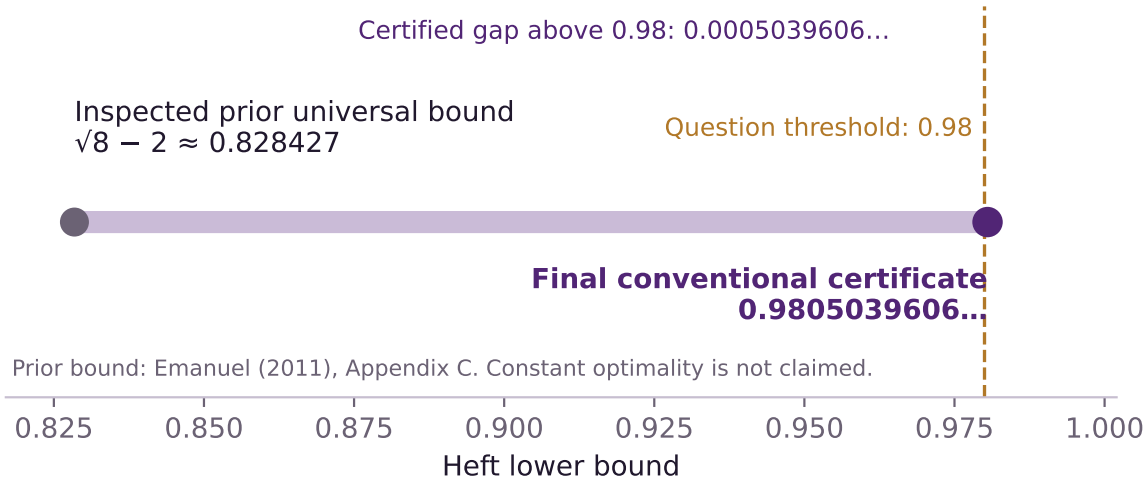


Figure 1: *The certified constant crosses the requested threshold. This is a comparison with the strongest general bound located in the inspected primary sources, not a universal priority or optimality claim.*

Discovering the opportunity

Purpose before application

The motivating opportunity was to make a consequential unanswered question yield an inspectable answer under available digital resources. The study did not begin with a prescribed customer workflow. It searched for targets with an external specification, an identifiable limitation, a strong conventional challenge, a plausible beneficiary and a decisive experiment. Mathematical construction became a viable lane because its first useful output can be checked digitally and its acceptance need not depend on a physical prototype or an unavailable archive.

For the selected problem, the limitation statement is:

> Existing finite enumeration and finite integer-programming searches that find no qualifying cover cannot rule out one at a larger interval length: a cutoff leaves infinitely many lengths unresolved. An explicit universal weight certificate resolves that remaining possibility while preserving complete classes, exact partitioning and minimum class size. A finite search could, of course, settle existence by finding a valid witness.

This statement concerns the insufficiency of a finite search as a universal proof. It does not claim that conventional mathematics, an existing university or a research organization lacks the ability to discover such a proof. Duality, polynomial approximation and Bernstein verification are established tools. The scientific contribution is the explicit witness and resulting theorem.

The immediate beneficiary is the research community studying covering systems. A separate possible buyer could commission reproducibility, qualified review or integration of a fixed certificate. The former is a demonstrated intellectual use; the latter is a commercial hypothesis. We did not establish that the theorem has a near-term industrial application or that a particular organization is unable to perform this work.

A small, diverse screen and its conventional challenges

Six directions formed the initial screen. The detailed limitation statements and primary-source comparisons are in the evidence package.

Decision-directed experimental instruments. The proposed instrument would choose tests that settle a useful property without identifying an entire model. Decision Region Determination and Hyperedge Cutting already target that distinction, while exact dynamic programming is practical on small finite cases (Javdani et al. 2014). The generic mechanism did not survive the prior-art challenge.

Combining incomplete causal evidence. The proposed representation would return sharp intervention bounds and extremal compatible worlds from partial observations. Automated partial identification already supplies this core mechanism (Duarte et al. 2024). A convenient compiler alone would not establish a new productive possibility.

Combining specialists without training data. The proposal sought transferable capability by aligning and merging networks. Data-free, cycle-consistent model merging is direct prior art (Crisostomi et al. 2024). A new label or a successful merge would not establish economic Alpha or a stronger accepted outcome.

Runtime enforcement. The proposed component would preserve useful action while blocking violations of a formal specification. Shield synthesis is an established construction (Bloem et al. 2015). A new implementation was insufficient for the flagship criterion.

Exact zero-error communication codes. This lane had a concrete improvement target and independent digital verification. Strong published witnesses, MILP, local search and straightforward compositions were admitted. The executed searches reproduced known constructions but did not improve them. We stopped the record-improvement claim rather than changing the channel or weakening the comparator (Itty et al. 2026; Buys, Polak, and Zuiddam 2026; Tandon 2026).

Finite residue partitions. The public question supplied a fixed threshold and clear original requirements. Finite exact-cover search was attempted first. Its failure to find a witness suggested, without proving, a lower-bound direction. A certificate that applies uniformly to all lengths became the surviving research target (Myerson 2026).

Additional scouting considered quantum-code and small-design records. Those screens did not produce a new accepted witness and are preserved as exploratory notes. Screening conclusions are literature judgments, not executed capability benchmarks.

Why this became feasible in this study

The changed arrangement combines an accessible, recently refreshed public question; explicit computational definitions and prior examples; interactive frontier-model reasoning; inexpensive standard optimization; and exact independent verification. The 2026 public updates make the outstanding threshold easy to identify without inventing a benchmark. They do not make the mathematical question new (Myerson 2026). No unique instrument unavailable to other organizations is claimed.

The useful representational move is from "find a cover at larger n" to "find one weight function that prices every permissible class." A finite search can miss a distant witness. A valid universal weight converts the same original question into finitely checkable polynomial obligations plus a mathematical tail argument. This is a change in what the particular completed artifact establishes, not a claim that dual certificates were invented here.

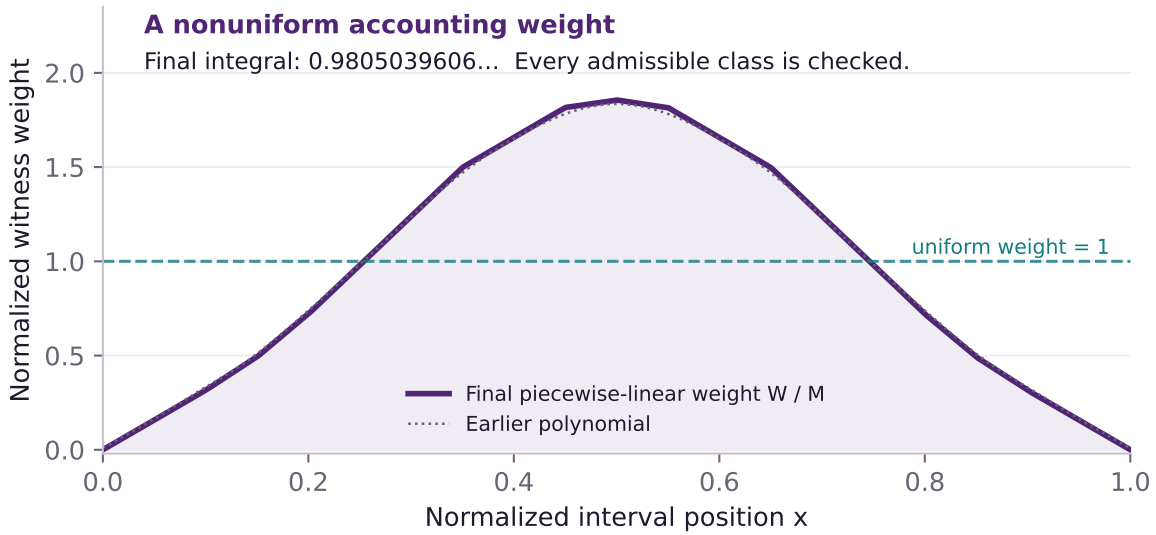


Figure 2: *The final piecewise-linear weight divided by its exact class maximum, with the earlier polynomial shown for comparison. These curves explain the construction; sampled plotting values do not certify it.*

The construction and its verification

A weight that every class must respect

Let W be the continuous piecewise-linear function with knots $r/20$, $r = 0, \dots, 20$. Its first eleven heights are

$$0, 159, 316, 493, 721, 981, 1240, 1500, 1658, 1818, 1858,$$

and the rest are their reflection about the midpoint. Extend it by zero outside the interval. Its exact integral is $A = 1963/2$. The construction proves that every complete progression with spacing d and at least two points in $[0,1]$ obeys

$$d \sum_j W(a + jd) \le M := \frac{18258529}{18240}.$$

Assign integer i the **cell integral** $w_i = \int_{(i-1)/n}^{i/n} W(x)\,dx$. Integrating the progression inequality over the phase within each cell proves $m \sum_{i \in C} w_i \le M$ for every complete restricted class C of modulus m. Since the classes partition all integers exactly once, $A \le Mh$. Thus $h \ge A/M = J$. This uses actual finite cells; it introduces no limiting approximation or exception for small lengths.

For a class with k points, spacing d and first point a, the complete-class parameter domain is the quadrilateral

$$0 \le a \le d, \qquad a + (k-1)d \le 1 \le a + kd.$$

Knot-crossing lines partition that quadrilateral into cells. Within a cell, the class sum has the form $A_1 ad + B_1 d^2 + C_1 d$. A maximum occurs on a cell boundary: a nonzero mixed coefficient makes the Hessian indefinite; when it is zero, the value is independent of a and extends to a boundary. On each boundary segment, exact quadratic maximization checks endpoints and any interior stationary point. All knot lines and all four exterior boundaries are included.

The primary conventional checker examines 1,967 segments for $k = 2, \ldots, 20$ and obtains exact maximum M. A separately written checker reconstructs the clipping from polygon-edge intersections and uses direct interpolation to fit the quadratic on every segment. It agrees. For $k \geq 21$, the total variation of the derivative of the zero extension is 21,200, yielding the universal tail bound $7905/8 < M$. Appendix A gives the full proof.

Independent checking and fresh confirmation

The final certificate therefore rests on two distinct exact implementations, a boundary-maximization lemma, an analytic tail and a cell-average transfer argument. Independence means separate implementations and derivations within the same AI-assisted study; it is not independent institutional or human peer review. The accepted relation at J is **greater than or equal to**. Strictness at that exact constant is not claimed; strictness above 0.98 follows because $J > 0.98$.

Obligation	Executed evidence	Meaning
Classes with 2–20 hits	All exact segment maxima agree in two implementations	Entire parameter domains, including all boundary edges
Classes with at least 21 hits	Exact tail $7905/8 < M$	No class-size cutoff remains
Every interval length	Exact cell-integral argument	No midpoint approximation or finite-n extrapolation
Full final wrapper	Both exact verifiers recomputed in 1.55 seconds in one local run	Local research verification, not a speed benchmark
Earlier polynomial certificate	78/78 patches in both implementations; exact all-length objective	A separate valid, weaker construction
Earlier fresh tests	512 classes, four new lengths, 19 scope decisions and 13 altered-input controls	Confirmation of the earlier artifact, not evidence silently reassigned to the final one

For the final piecewise-linear certificate, a new frozen seed (`202610274852`) generated 512 complete classes: 256 short classes and 256 longer classes. Direct phase evaluations and exact cell-integral inequalities all pass; cell weights telescope at nine fresh interval lengths. Two deliberately invalid mathematical variants are rejected. After correcting the JSON-null interface error, a separate final audit recomputed the stronger proof in 1.49 seconds, matched 22/22 scope decisions, and refused all 21 specified isolated CLI controls. One missing-helper case exits with an import error rather than structured refusal; it produces no acceptance. These cases are implementation checks, not a statistical substitute for proof or a model benchmark.

The final wrapper validates the requested scope, matches frozen source bytes and recomputes the proof in an isolated directory. It does not accept a result from an archived success flag. It refuses altered or missing proof inputs, stronger unsupported bounds and strictness at the exact final constant. Its hashes are reproducibility controls, not signatures proving scientific correctness or ownership. Both research wrappers refuse optimized Python because some underlying checkers use assertions.

The earlier polynomial remains useful independent evidence. Its exact coefficient file, both continuous-domain implementations, analytic tail and all-length identity are preserved in Appendix B and the package. Its bound $h > I = 0.9804028583156853\ldots$ is weaker than the final conventional certificate. Its 512 fresh class tests used seed `202610274850`, lengths up to one million and complete classes under a frozen protocol; new midpoint lengths were 251, 257, 509 and 601.

These finite checks confirm implementation behavior and do not substitute for either universal proof.

The failure that changed the result

The first degree-20 numerical candidate had integral approximately 0.98558247 and appeared stronger on a coarse grid. It was invalid. Continuous exploration exposed a violation; independent exact checking found an integer realization at $n = 26$, modulus 15, residue 11, with class $\{11, 26\}$. Its normalized constraint value is approximately $1.046411318 > 1$. That candidate is rejected in the publication and remains in the evidence package.

Dense constraints and eight cutting-plane iterations produced the replacement. Decimal coefficients were converted to exact rationals and scaled by $2499/2500$ before certification. The frozen JSON still says `UNVERIFIED`, reflecting its state when selected; later acceptance receipts establish verification without rewriting the candidate's identity. Two implementation errors in the all-length refinement were also caught: accidental floor division of rational expressions, and an attempted positivity interval larger than the discrete-length argument required. Both original attempts and their corrections are retained. Final adverse testing also caught an interface error: an explicitly supplied JSON-null request was treated as an omitted request. The mathematical proof still ran correctly, but the schema refusal was wrong. The final interface was corrected and refrozen; the failing code and receipt remain archived.

What the conventional challenge does and does not show

Nineteen conventional finite-cover runs produced seventeen valid incumbents and no cover below 0.98; their aggregate recorded wall time was 146.46 seconds. Two runs ended without an incumbent. A feasible incumbent is not an optimum certificate, and none of these finite searches establishes an all-length impossibility result.

The execution comparison gives ordinary methods the discovered formulation, exact witness, proof and checker. They can reproduce the same accepted result. This equality prevents an exclusive solver claim. It does not erase the mathematical consequence of a previously unrecorded explicit certificate.

The strongest conventional challenge succeeds. The fresh workstream received the public question, primary references, ordinary libraries and the same available tool permissions, without seeing the polynomial witness. Standard finite LP duals suggested its piecewise-linear construction. Its final exact certificate was completed at 22:23:45 UTC, 4 minutes 36 seconds after the recorded start. Recorded deliberate numerical work was approximately 145.9 CPU seconds, plus an interrupted setup lasting less than six wall seconds whose CPU usage was not captured.

The challenger initially used the convenient scale $2003/2$, proving $h \geq 1963/2003 > 0.98$. Its own exact maximum was smaller. Independent audit used that maximum without changing any heights or useful requirements, yielding the stronger J. Reporting only the looser scale would understate the conventional result. The conventional construction meets the original target and exceeds the first polynomial bound. We therefore stop any claim of exclusive superiority for that first approach and publish the strongest joint scientific result.

This is a new explicit certificate relative to the inspected prior sources, produced using established techniques. It does not establish a new optimization method, an advantage from specialist retention, or a reason an existing organization could not produce it.

Discovery must be assessed separately. The main opportunity search was adaptive and exploratory. It was not a randomized, identical-information comparison of ordinary and expert model use. A targeted fresh-context conventional challenge was launched after the mathematical question had been selected, with a ten-minute wall ceiling and 300 CPU-second numerical-search ceiling. Its initial information and search are retained separately. That experiment cannot establish population-level discovery advantage or prove that humans or ordinary agents could not find the result.

Building an economic possibility around a useful artifact

Actual evidence and the proposed service

The executed deliverable is a proof package. Actual customers, recognized revenue, receipts and deployed customer acceptance are all zero. Complete actual cost is unknown because full model, authoring, labor and infrastructure bills were not supplied. Therefore actual operating profit and full-project profit are unknown, rather than zero or a modeled positive number.

The planning model prices a possible service around **this existing certificate**: reproduction, qualified mathematical review, source attribution, packaging, integration into a beneficiary's agreed environment, correction and one year of response and maintenance. It does not promise a newly solved open problem for each customer. Research groups, foundations, mathematical-software organizations and publishers are possible counterparties; none was contacted.

There is no observed market price for this exact offering. Human-led development and delivery resources are estimated explicitly, as permitted planning assumptions. They are not customer value, willingness to pay, observed AI costs or a wage-derived consulting quote. Fees are tested inputs. Any value-based proposal must be bounded by a buyer's actual avoided cost, accepted scientific need and willingness to commit; those quantities remain unmeasured.

The central estimate includes 110 fixed human-resource hours at assumed rates of USD90–140 per hour, 20% overhead and USD360 fixed compute/storage allowances. It includes unsuccessful discovery, construction, verification, correction, oversight, acquisition, coordination and shared maintenance. Each beneficiary adds 10.5 hours plus overhead and a USD40 compute/distribution allowance. Development is charged once; later retention would incur only additional attributable work. Appendix D lists the activities.

Complete profit, funding and adverse outcomes

Let F be fixed full-project cost, v the marginal delivery cost, p the fee, q the processing fraction and N the number of accepted, paying beneficiaries. The modeled full-project profit is

$$\Pi = N\{p(1-q) - v\} - F.$$

In the central scenario, $F = 15{,}168$, $v = 1{,}426$ and $q = 0.03$. Four prospective sponsors at an assumed USD6,000 each yield USD24,000 revenue and USD21,592 complete project cost. The full-project profit is USD2,408 before financing and tax. Delivery operating surplus before the once-counted USD5,844 development allocation is USD8,252. Neither number is an observed commercial result.

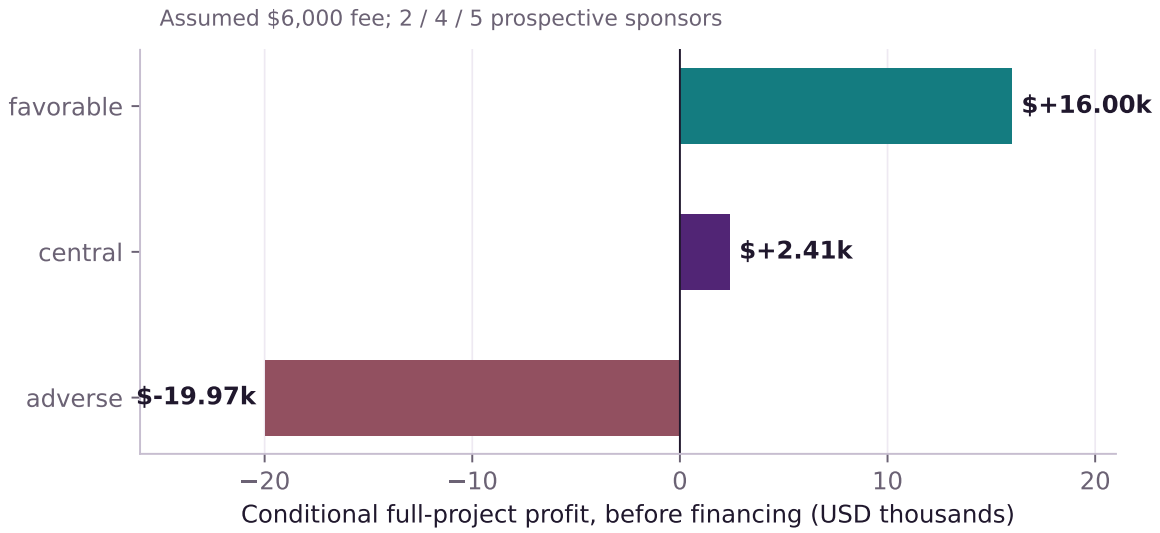


Figure 3: *Conditional full-project results. The central profit is modest and depends on four paying sponsors; the adverse loss remains visible. Actual revenue is zero and actual complete profit is unknown.*

Conditional scenario	Sponsors	Revenue	Complete cost	Project profit	Added funding need
Favorable	5	$30,000	$14,001.73	$15,998.27	$2,202.50
Central	4	$24,000	$21,592.00	$2,408.00	$6,573.20
Adverse	2	$12,000	$31,968.80	−**$19,968.80**	$18,880.80

All scenarios assume USD6,000 per sponsor and USD5,000 initial unrestricted capital. The favorable case uses 65% of central hours and 90% of central rates; the adverse case uses 160% of hours and 110% of rates. These are deterministic sensitivities, not empirical distributions. The adverse case is economically losing and cannot repay the illustrated bridge from project receipts and original capital. It must not be described as merely underfunded.

At four sponsors, the central break-even fee is USD5,379.38 each; at USD6,000, four accepted payers are required. A fixed-assumption grid varies one to twelve participants and eight fees. A participation-sensitive schedule tests USD18,000 for one sponsor, USD10,000 for two, USD6,000 for four, USD4,800 for six and USD4,000 for eight. These conditional discounts reflect shared fixed costs; they are neither a proven optimum nor evidence that participation rises at those prices. An exclusivity requirement, incompatible integration environment or unpaid sponsor changes the economics. A separate 60-case value-pricing grid assumes a buyer value already net of internal costs and incremental to the free public-artifact option, then tests capture fractions of 25%, 50% and 75%. At four buyers and 50% capture, the necessary net value for break-even is USD10,758.76 each. Hypothetical value USD12,000 gives the USD6,000 central fee and USD2,408 project profit; value USD6,000 gives a USD3,000 fee and USD9,232 loss. These are required-value sensitivities, not measured buyer value or uptake. Fee below value is only a necessary participation condition, not proof of demand.

The cash model uses a refundable 40% deposit, a 30% payment at accepted core delivery and 30% at final acceptance. Deposits remain restricted until earned. The central arrangement reaches an additional unrestricted cash requirement of USD6,573.20 above the unchanged USD5,000 capital. The plotted negative balance is a counterfactual stop point: work cannot continue unfunded, and refundable deposits cannot fill the gap.

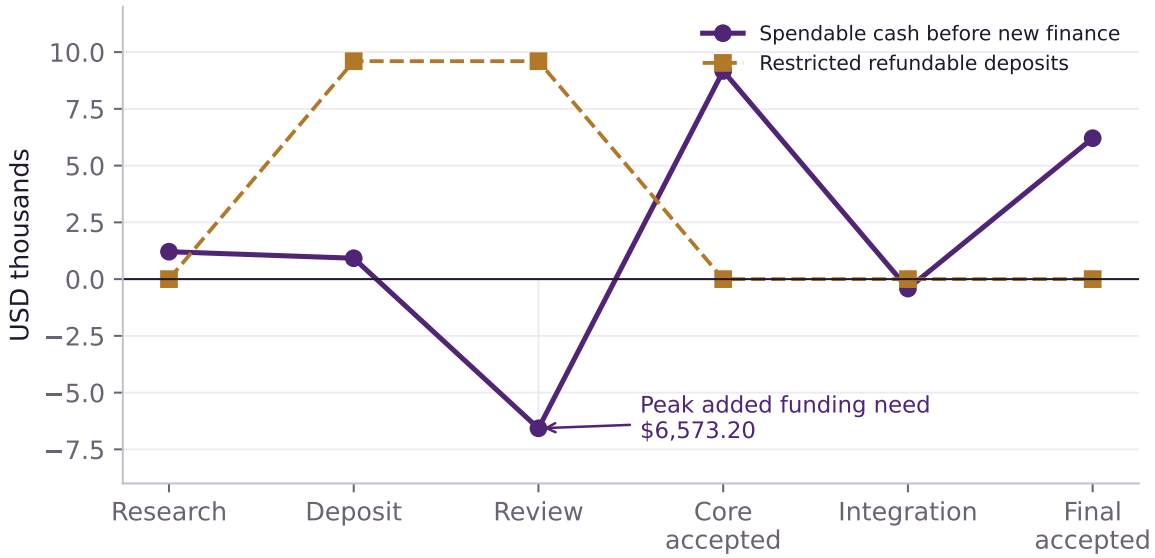


Figure 4: *Central cash timing before new finance. Refundable deposits remain restricted. A positive project margin does not finance the review work that precedes acceptance.*

A hypothetical 90-day bridge at a 2% origination charge and 12% simple annual interest requires USD6,707.35 principal to provide the needed net proceeds. Its USD335.37 financing cost leaves conditional central profit of USD2,072.63. This is a financing sensitivity, not an available lender offer. Delayed acceptance, failed review or cancellation can make it inadequate. Before first acceptance, cancellation returns the USD9,600 deposit and leaves USD11,573.20 unrecovered central cost.

Expected support and maintenance are already costed and assumed prepaid at integration in the timing scenario. A remaining service obligation survives final delivery. A separate reserve of 5% of earned fees encumbers cash for adverse obligations; it is not a second expense. Financing principal, deposits, reserves, token issuance and token appreciation are not operating earnings. All figures are pre-tax planning values, with applicable pass-through taxes excluded from revenue. Exact-rational accounting and a separately written cash reconstruction agree.

Complementarity at enterprise level

The comparison includes the existing finite-search arrangement, technical proof alone, coordination alone, both together, and an equally capable conventional shared arrangement.

Arrangement	Universal mathematical deliverable	Commercial interpretation
Finite MILP alone	Not established	No sale credited for a universal answer
New certificate; ordinary four-client delivery	Established digitally	Conditional full-project profit of $2,408
Coordination of finite results	Not established	Shared payment does not supply the missing proof
Certificate; four compatible sponsors	Established digitally	Conditional full-project profit of $2,408

Conventional shared arrangement; same artifact	Same accepted result	Same conditional $2,408 profit

All accepted four-beneficiary arrangements hold buyer count, fees, access, work and cost assumptions fixed. Ordinary acquisition and coordination costs are included in each. They tie; no separate incremental coordination benefit or exclusive technical-commercial interaction is identified. A one-beneficiary case at the same assumed USD6,000 fee loses USD10,774, but that is a participation sensitivity, not a causal estimate that coordination creates three customers. The supported conditional mechanism is sharing fixed review and publication costs among compatible requirements. A numerical factorial interaction is not estimated because the finite-search arms do not deliver the same accepted product. Assigning them invented profitable sales would manufacture a comparison.

Participants would need compatible acceptance terms, nonexclusive access, clear rights to contributed code and an agreed allocation of review, correction and withdrawal costs. No exclusive ownership of a mathematical problem or theorem is sold. A sponsor can decline or use the public artifact independently. These outside options make willingness-to-pay validation essential. The arrangement would fail if shared review is unacceptable, environments diverge, promised human review cannot be staffed, or commitments do not become earned receipts.

A successor with a useful inheritance

Retention is selective

The successor inherits both exact witnesses, their readable proofs, independent checkers, scope rules, counterexamples and economic boundaries. That is technical portability. It is not by itself net Alpha.

Fresh scope tasks cover translated integer intervals, repeated moduli, weaker thresholds and unsupported requests such as singletons, two dimensions or a stronger bound. A competent recipient with the same ordinary-library artifact has the same mathematical information. The recorded scope decisions confirm supported reuse and refusal behavior for their identified artifact; they are not an LLM performance comparison. No incremental task benefit over that ordinary library was measured.

Additional specialist packaging is modeled at USD1,042: USD330 extraction, USD280 extra validation, USD220 integration, USD200 upkeep and USD12 storage. The planning model admits no unsupported incremental benefit, so paid extraction is **HOLD**. Zero admitted benefit is not a measured zero effect or a theorem that future benefit is impossible. The existing research artifact remains useful without that expenditure. Gym practice and further specialist training require a fresh target and an economic reason; they are not automatic reinvestment milestones.

Frontier-model operation and its measurement boundary

The live authoring session identifies GPT-6 Astra. Official documentation inspected on the study date lists identifier `gpt-6-astra`, a 1,050,000-token context window, a 128,000-token output limit, tool use and configurable reasoning effort. Documented standard short-context prices are USD10 per million input tokens, USD1 cached input and USD50 output; long-context and faster modes have different rates (OpenAI 2026b, 2026a). These are verified planning references, not a bill for

this session.

Work was decomposed into opportunity search, conventional challenge, source/definition audit, construction, independent checking, economic reconciliation and publication. Recovery included rejecting an attractive infeasible polynomial and preserving corrected code. Those are descriptions of expert operation. No local API credential, metered provider call log, precise sampling configuration or complete authoring-token bill was available. We therefore ran narrower executable mathematical and accounting tests, not a simulated model-capability measurement.

Operational Alpha would require a fresh, matched comparison of ordinary and expert operation of the same model, with accepted outcomes, errors, correction, human effort and resource cost all counted. This study does not supply that comparison. Here the conventional challenger also independently discovered a successful formulation for the selected problem. That substantive success reinforces the absence of an exclusive capability claim. A broader discovery comparison still needs matched initial information, multiple fresh tasks and complete costs.

Existing ecosystem components, actually exercised

The study continues the inspected repository and issue history. It does not replace the active execution, specialist or enterprise teams. The native source closure contains 44 unchanged files pinned to commit `47c17f6`, with byte equality checked against the inspected main source. Separate native receipts bind the earlier polynomial artifact and the final stronger certificate, including its corrected wrapper and fresh confirmation hashes. An existing archive-study adapter is reused unchanged (MONTREAL.AI 2026a, 2026b).

Insight produced a neutral single-candidate review dossier. It records the selected result and checks its local structure; supplied neutral scores do not measure opportunity discovery. **NovaSeed** records an unminted, unfunded research draft with provenance. **FusionPlan** compiles and verifies four job commitments, with altered-plan and altered-dossier refusals. A verified plan does not assert that downstream jobs were executed. The existing production capsule refuses the unsupported scientific scope.

The local Journal, Trust and CostLedger fixtures record four events and reject a complete-cost seal while authoring costs are unknown. Profit remains null. Temporary fixture identities are explicit identity simulation, not live `*.agent.agi.eth` authentication. MARK contracts and the review pathway were inspected, but MARK was not activated. No minting, trading, spending, permission change, customer contact or token operation occurred. USDC settlement and applicable $AGIALPHA utility remain distinct; neither contributes imaginary revenue.

Cryptographic hashes protect the integrity of recorded content. They do not establish scientific accuracy, ownership of an opportunity, foresight or investment value. Publication is a research-review handoff through the existing repository process, separate from a qualified production release.

Learning from the existing evidence

Three foundations guide the study. *Transformational Intelligence* separates purpose, formulation and evaluation, and explicitly admits an unrestricted optimal conventional planner. *Invent the Economy* distinguishes technical possibility, complete profitability and financing; a positive project with insufficient prefunding differs from one whose added costs make its profit negative. *Transformational Autonomous Enterprises* requires mission-specific revalidation and comparison with a competent

direct specialist, rather than counting graph growth or imports as Alpha (MONTREAL.AI 2026g, 2026d, 2026f).

The recent event-history construction tied classical parity and closure. The archive instrument made valid exact inferences but tied conventional methods; its corrected economics and unresolved successor questions prevented a flagship claim. The disclosure follow-up matched established auditing and remained losing in its tested complete scenarios (MONTREAL.AI 2026c, 2026b, 2026e). These are different offerings; their profits and losses are not imported into this study.

The requested DNA-encoding lesson is retained as user-supplied prior context: a useful construction was matched conventionally. Targeted source and file searches did not resolve its exact artifact. We therefore do not invent its case counts, financial figures or a source path. This unresolved lineage item does not support a positive claim here.

The practical correction is earlier challenge. Broad mechanisms already supplied by prior art were stopped before implementation. Known code records were preserved as strong baselines. The infeasible polynomial was rejected despite its attractive numerical integral. The final paper leads with the checked theorem while keeping commercial and comparative limits beside it.

Three horizons, with explicit dependencies

Near term — a durable mathematical result. The completed capability is a locally reproducible, exact answer to the stated covering question, with two implementations, fresh checks, a refusal boundary and a retained failure history. Its scientific significance is a stronger universal bound and a reusable proof artifact. Its actual commercial return is unmeasured. External expert review of definitions, mathematics and priority is the smallest decisive next scientific step.

Expansion — a portfolio of accepted certificates. Related finite-cover parameters, sharper constants and other discrete construction problems are plausible fresh work. Each needs a fixed externally meaningful target, unrestricted conventional search, a proof or independently accepted witness, full costs and a refusal if the target is missed. A certificate-review service additionally needs a real sponsor's acceptance criteria, resource quote, willingness to pay and financing commitment. Failure of external review, conventional equivalence without separately demonstrated discovery value, or insufficient paid demand falsifies the respective expansion claim.

Long term — productive infrastructure for discovery. The ambition is an enterprise that builds instruments, representations and evidence others can use to undertake work previously unsupported by their available artifacts. A reliable portfolio could support new research collaborations; validated delivery and transferable capability could support broader instruments. Repeated useful discoveries, independent acceptance, economic surplus and selective retention are separate causal steps. None implies inevitable acceleration, a measured enterprise valuation or ownership of a future industry.

The successor receives something concrete: a problem boundary it no longer needs to guess, the means to verify that boundary, and an honest record of where the result stops. **Find your Alpha. Build your SUCCESSOR Ω.**

Appendix A. A universal piecewise-linear weight certificate

A.1 Statement and exact coefficients

Let finitely many complete residue classes, restricted to the integers $1, \dots, n$, partition that interval. Suppose each restriction contains at least two integers. If their moduli are $m_1, \dots, m_r$, then

$$\boxed{h = \sum_{i=1}^{r} \frac{1}{m_i} \geq J := \frac{17902560}{18258529} = 0.9805039606421744\ldots > \frac{49}{50}.}$$

The proof permits repeated moduli and therefore answers the original distinct-modulus question. It retains complete restrictions, exact partitioning and the minimum of two interval elements per class. No optimality assertion accompanies the constant J.

Define W as the continuous linear interpolant at the knots $j/20$, for $0 \leq j \leq 20$, with $W(1-x) = W(x)$ and the following integer coefficients. Extend W by zero outside $[0,1]$.

j	$W(j/20)$
0	0
1	159
2	316
3	493
4	721
5	981
6	1240
7	1500
8	1658
9	1818
10	1858

Reflection supplies the remaining ten coefficients. Exact trapezoid integration and summation of derivative jumps give

$$I_W := \int_0^1 W(x)\,dx = \frac{1963}{2}, \qquad V := \mathrm{TV}(W') = 21200.$$

Here W' denotes the derivative of the zero extension. Its variation includes the jumps from zero derivative at both support endpoints. The normalization used below is

$$M = \frac{18258529}{18240}, \qquad J = \frac{I_W}{M}.$$

A.2 The complete progression domain

For $k \geq 2$, consider the sum

$$S_k(a,d) = d \sum_{\ell=0}^{k-1} W(a+\ell d), \qquad d > 0.$$

The closed domain

$$D_k = \{(a,d) : 0 \leq a \leq d, \quad a+(k-1)d \leq 1 \leq a+kd\}$$

contains every complete k-point progression in the open unit interval. The inequalities allow its predecessor or successor to lie at an endpoint; this enlargement is harmless because $W(0) = W(1) = 0$. In particular,

$$\frac{1}{k+1} \leq d \leq \frac{1}{k-1}.$$

The domain is the compact quadrilateral with consecutive vertices

$$\left(\frac{1}{k+1}, \frac{1}{k+1}\right), \quad \left(\frac{1}{k}, \frac{1}{k}\right), \quad \left(0, \frac{1}{k-1}\right), \quad \left(0, \frac{1}{k}\right).$$

We prove $S_k \leq M$ throughout D_k for every $k \geq 2$.

A.3 Exact maximization for short progressions

For each $2 \leq k \leq 20$, subdivide D_k by all knot-crossing lines

$$a + jd = \frac{r}{20}, \qquad 0 \leq j < k, \quad 0 \leq r \leq 20.$$

On each resulting polygonal cell, the arguments of W stay in fixed linear pieces. Consequently

$$S_k(a,d) = Aad + Bd^2 + Cd$$

for constant rational coefficients. If $A \neq 0$, its Hessian has determinant $-A^2 < 0$, excluding an interior local maximum. Equivalently, its derivative in a is the nonzero quantity Ad. If $A = 0$, the objective is independent of a: a maximizing section at fixed d reaches the cell boundary with the same value. Degenerate cells already belong to boundaries. Thus a global maximum occurs on an enumerated boundary, including possible stationary ridges.

Every boundary is a knot-crossing line or one of the four edges of D_k. The lines with $(j,r) = (0,0)$ and $(k-1,20)$ cover $a = 0$ and $a + (k-1)d = 1$. The remaining edges are included explicitly as $(j,r) = (-1,0)$ and $(k,20)$, representing $a = d$ and $a + kd = 1$.

On a fixed line $a + jd = r/20$, every crossing with another knot line occurs at

$$d = \frac{s-r}{20(\ell-j)}, \qquad \ell \neq j.$$

The domain endpoints and these crossings divide the line into finitely many rational segments. On

each, $S_k = \alpha d^2 + \beta d$. Its maximum occurs at an endpoint or, when $\alpha < 0$, at the stationary point $-\beta/(2\alpha)$ if that point belongs to the segment. Degenerate point segments are also evaluated.

The standard-library checker `fresh_conventional/verify_weight.py` implements this enumeration using exact integers and fractions. All nineteen cardinalities pass, and their largest value is precisely

$$M = \frac{18258529}{18240},$$

attained at $k = 2$, $a = 7/20$ and $d = 4273/9120$. The original enumeration checks 1,967 segments, including point cases and repeated descriptions.

The independent checker `lineage/audit_conventional_weight.py` reconstructs line segments by intersecting polygon edges, rather than by the first checker's inequality-clipping formulas. It interpolates W directly, reconstructs each quadratic from exact endpoint values, checks its midpoint identity and evaluates stationary points directly. It reproduces every per-cardinality maximum. Its different bookkeeping gives 1,863 positive-length segments and 360 interior stationary evaluations, with degenerate points checked separately. The audit receipt is `confirmation/conventional_independent/INDEPENDENT_WEIGHT_AUDIT.json`. These are separate implementations within one study, not external peer review or proof-assistant verification.

A.4 An analytic bound for all remaining cardinalities

For a continuous compactly supported piecewise-linear function, the composite trapezoid estimate on a lattice of spacing d is

$$\left| d \sum_{z \in \mathbb{Z}} W(a + zd) - \int_{\mathbb{R}} W(x)\, dx \right| \le \frac{d^2}{8} \operatorname{TV}(W').$$

Indeed, on a lattice cell $[u, u + d]$, the trapezoid error is the integral of $(x - u)(u + d - x)/2$ against the signed measure of derivative jumps. This kernel has magnitude at most $d^2/8$. Summing over cells and using the measure's total variation proves the estimate. Endpoint jumps of the zero extension are included, so cells crossing the support boundary require no exception.

For $(a, d) \in D_k$, the listed progression contains every nonzero contribution of the lattice. When $k \ge 21$, the bound $d \le 1/20$ therefore gives

$$S_k(a, d) \le \frac{1963}{2} + \frac{21200}{8 \cdot 20^2} = \frac{7905}{8} < M.$$

Together with the finite exact computation, this proves $S_k \le M$ for every admissible progression, without a cardinality cutoff.

A.5 Exact transfer to every integer interval

For an arbitrary partition of $\{1, \dots, n\}$, define cell-integral weights

$$w_j = \int_{(j-1)/n}^{j/n} W(x)\, dx.$$

Consider a complete restricted residue class C modulo m. For each $0 < t < 1$, its points

$$x_j(t) = \frac{j-1+t}{n}, \qquad j \in C,$$

form a progression of spacing $d = m/n$ with at least two members. Its predecessor integer is at most zero, so its normalized predecessor is at most $(-1+t)/n < 0$. Its successor integer is at least $n+1$, so its normalized successor is at least $1 + t/n > 1$. The progression thus belongs to $D_{|C|}$.

Integrating its certified inequality in t yields

$$\frac{m}{n} \sum_{j \in C} \int_0^1 W\left(\frac{j-1+t}{n}\right) dt = m \sum_{j \in C} w_j \le M.$$

Sum over the partition. Every integer cell occurs once, giving $I_W \le Mh$ and hence $h \ge J$. This applies directly to every admissible n; no limiting argument, midpoint correction or unverified small-length case remains. For $n = 1$ the hypotheses are impossible.

A.6 Normalization and contribution

The discovery report used the convenient safe scale $T = 2003/2$, obtaining $h \ge 1963/2003 > 0.98$. Its computation already established the smaller exact maximum M. Normalizing by M gives the stronger bound J without changing the knots, domain or proof obligations. The theorem states $h \ge J$; it does not assert strictness at J.

Discovery used standard linear and mixed-integer optimization, followed by exact rational geometry and a classical quadrature estimate. The contribution is the explicit certificate and resulting universal bound. It establishes neither a new general solver nor an exclusive AI capability or controlled discovery advantage.

Appendix B. The earlier polynomial proof

B.1 Definition and exact witness

Let a complete residue class modulo a positive integer m be restricted to $\{1, \dots, n\}$. Suppose finitely many such restrictions partition that interval and each contains at least two points. Repeated moduli are allowed in this proof. Let h be the sum of reciprocal moduli.

The coefficient vector, in increasing even Chebyshev order, is

$$\begin{aligned}(b_0, \dots, b_{10}) = (&7061527976, -8616261687, 2079655784,\\ &-480883959, 30533879, 8370806, 32646497,\\ &-16295510, 3707761, -15109523, 30943868).\end{aligned}$$

Let T_r denote the Chebyshev polynomial of the first kind. Define

$$f(x) = \frac{2499}{2500 \cdot 10^{10}} \sum_{j=0}^{10} b_j T_{2j}(2x-1).$$

It satisfies $f(1-x) = f(x)$. Each basis element has integral $(1-4j^2)^{-1}$ on $[0,1]$, so the exact witness integral is

$$I = \frac{1891871140656049}{1929687500000000} = 0.9804028583156853\ldots.$$

B.2 The class domain and the dual argument

A class has interval hits $r, r+m, \ldots, r+(k-1)m$, where $k \geq 2$. Put $d = m/n$ and $a = (r-1/2)/n$. The previous and next members lie outside the interval. Consequently

$$\max(0, 1-kd) \leq a \leq \min(d, 1-(k-1)d), \qquad \frac{1}{k+1} \leq d \leq \frac{1}{k-1}.$$

These closed inequalities relax the strict exterior conditions of actual integer classes and thus enlarge the checked domain. Define the class constraint

$$d \sum_{j=0}^{k-1} f(a+jd) \leq 1. \tag{B1}$$

Assign integer i the weight $f((i-1/2)/n)/n$. If (B1) holds, the class's assigned weight is at most $1/m$. Exact partitioning then yields

$$h \geq M_n := \frac{1}{n} \sum_{r=1}^{n} f((r-1/2)/n). \tag{B2}$$

There is no sign restriction on individual weights in this summation argument. We next establish (B1) for every class and $M_n > I$ for every $n \geq 2$.

B.3 Every short class: 78 exact continuous certificates

Write $F(z) = f((z+1)/2)$ and $u = 2a + (k-1)d - 1$. Reflection of the class allows $u \geq 0$ because F is even. The admissible domain becomes

$$0 \leq u \leq \min((k+1)d - 1, 1-(k-1)d).$$

For $(s, v) \in [0,1]^2$, its two halves are parameterized by

$$\begin{aligned} &\text{lower spacing:} \quad d = (k+s)/(k(k+1)), \qquad u = vs/k, \\ &\text{upper spacing:} \quad d = (k-1+s)/(k(k-1)), \quad u = v(1-s)/k. \end{aligned}$$

For each $k = 2, \ldots, 40$ and each half, the polynomial

$$P_k(s, v) = 1 - d \sum_{j=0}^{k-1} F\big(u + (2j-k+1)d\big)$$

has bidegree at most $(21, 20)$. Clear its positive rational denominator and transform to the tensor Bernstein basis. A polynomial in that basis is a convex combination of its coefficients on the unit square. Nonnegative coefficients therefore prove $P_k \geq 0$ on that square. Exact de Casteljau midpoint subdivision applies the same test on smaller rectangles whose union is the original square.

The primary checker uses the identity

$$\sum_{j=0}^{k-1}(u+(2j-k+1)d)^{\ell}=\sum_{r=0}^{\ell}\binom{\ell}{r}u^{\ell-r}d^{r}\sum_{j=0}^{k-1}(2j-k+1)^{r}.$$

Symmetry eliminates the odd terms. After clearing the coefficient denominator, a common denominator contains $[k(k\pm1)]^{21}$ and is positive. All 78 patches pass: 106 visited nodes, 92 accepted leaves and maximum subdivision depth six. The independent sparse-Horner implementation checks the same polynomials with a different conversion and subdivision sequence: 95 accepted leaves and the same maximum depth. Exact leaf rectangles and bounds are retained. Different trees do not change the covered domain.

B.4 Every larger class: an analytic tail

Define $g=f-f(0)$ on $[0,1]$ and extend it by zero to the real line. Symmetry makes this extension continuous. Exact univariate Bernstein bounds on 64 rational subintervals give

$$|f(0)|<3/250,\qquad \|f''\|_{\infty}<486,\qquad \mathrm{TV}(g')<24.$$

The variation includes both endpoint jumps of the extended derivative. Its tighter computed upper bound is approximately 23.05544856893; the second-derivative bound is approximately 485.76248165584. Both arise from rational inequalities, not rounded acceptance thresholds.

For a continuous compactly supported function whose derivative has bounded variation, the shifted-grid trapezoid bound is

$$\left|d\sum_{j\in\mathbb{Z}}g(a+jd)-\int_{\mathbb{R}}g(x)\,dx\right|\le\frac{d^2}{8}\,\mathrm{TV}(g').$$

To obtain the constant, integrate the piecewise-linear interpolation error on each grid cell against the distributional second derivative. The corresponding kernel has magnitude at most $d^2/8$; summing the variation over cells gives the bound. The integral of the interpolant equals the composite trapezoid sum. Zero extension and continuity account for cells crossing the endpoints.

For a complete class, $|kd-1|\le d$. Adding back $f(0)$ gives

$$d\sum_{j=0}^{k-1}f(a+jd)\le I+|f(0)|d+\frac{d^2}{8}\,\mathrm{TV}(g').$$

For $k\ge41$, $d\le1/40$, so the right side is at most

$$I+\frac{3}{10000}+\frac{3}{1600}=I+0.002175<1.$$

Thus (B1) holds for every larger class as well. The independent tighter rational bound is approximately 0.98250103613, also strictly below one.

B.5 Every interval length: the strict objective

Let $t = 1/n^2$, $z_i = (2i - 1 - n)/n$ and $A_r(t) = n^{-1} \sum_i z_i^{2r}$. The identity

$$\sum_{i=1}^{n} \left((z_i + 1/n)^{2r+1} - (z_i - 1/n)^{2r+1} \right) = 2$$

telescopes. Binomial expansion gives

$$A_0(t) = 1, \qquad A_r(t) = \frac{1 - \sum_{\ell=1}^{r} \binom{2r+1}{2\ell+1} t^\ell A_{r-\ell}(t)}{2r+1}.$$

Writing $F(z) = \sum_{r=0}^{10} c_r z^{2r}$, this yields the exact identity

$$M_n = \sum_{r=0}^{10} c_r A_r(t) = I + tQ(t),$$

where Q has degree nine. The coefficient list agrees exactly with a separate Euler–Maclaurin derivation. The independent checker uses an explicit Chebyshev coefficient formula and the telescoping recurrence, without importing the first derivation.

All Bernstein coefficients of Q are strictly positive on each of the 16 intervals $[j/256, (j+1)/256]$, $j = 0, \ldots, 15$. Hence $Q(t) > 0$ on $[0, 1/16]$, which covers every $n \geq 4$. The two remaining exact differences are

$$M_2 - I = \frac{184569613348897}{61750000000000000} > 0,$$

$$M_3 - I = \frac{10332365069712591792733 63}{53827234190437500000000000} > 0.$$

Therefore $M_n > I$ for every $n \geq 2$. Together with (B2), this proves $h > I$. The JSON evidence contains every rational coefficient and each positive interval bound.

A redundant weaker completion also survives: the midpoint-remainder inequality gives $M_n > I - 486/(24n^2) > 0.98$ for $n \geq 225$, with smaller n checked exactly. This overlap provides an additional check on the original threshold conclusion; it is not needed for the stronger constant.

Appendix C. Study protocol, failures and comparison limits

C.1 Budget and chronology

The initial protocol was recorded with start time 21:57:28 UTC on 27 October 2026, a 120-minute wall ceiling, at most four concurrent workers, a 30 CPU-minute deliberate-search ceiling and a 1,300 MiB memory ceiling per search. It reserved at least 35 minutes for fresh confirmation, reconciliation and publication. No external spending was authorized or performed. Search, source inspection and publication are different resource categories. Complete aggregate CPU and peak-memory telemetry were not captured; the package records per-run limits and available elapsed/CPU observations rather than certifying an exact total. Complete model billing remains unavailable.

The first checkpoint was the diverse opportunity screen and strongest identified conventional challenges. The second was the working exact certificate and provisional decision. The polynomial witness and checker files were frozen for confirmation at 22:21:02 UTC. After the stronger conventional result passed independent audit, a separate final freeze was recorded at 22:30:41 UTC. An independent malformed-request check then found a JSON-null interface error; the corrected final wrapper received an additive second freeze before its final adverse suite. All snapshots are preserved. The supplemental exact-case plan was recorded before those fresh cases were executed. Numerical search configuration was consolidated after discovery and is explicitly retrospective; it is not presented as preregistration.

Early stop rules included a same-requirement conventional match, no independently checked improvement, resource exhaustion and unavailable authorized access. The zero-error lane was stopped without a record claim. The initial polynomial was rejected. The final mathematical claim survives the exact checks, while commercial validation, exclusive capability and discovery Alpha remain unestablished. Publication is bounded to this review edition; no continuous indefinite search is implied.

C.2 Preserved conventional results

The C_{15}^{3} code baseline contains 383 words, verified independently against all 73,153 word pairs and 3,064 permitted channel outputs. A 30-second unseeded global MILP found only 338, so it was never substituted for the stronger baseline. Two approximately 180-second seeded neighborhood searches retained 383. These are bounded search outcomes, not optimality or impossibility proofs.

The C_{7}^{5} reconfiguration search exhausted 243 states within its one-exchange class without finding a 368th word. Product/switch searches reproduced 134,753 words in C_{7}^{10} but did not improve the known construction. Subsequent primary-source inspection found stronger asymptotic recursion results, so older July bounds were not used as the final asymptotic comparator (Itty et al. 2026; Buys, Polak, and Zuiddam 2026; Tandon 2026). No physical-channel claim or market price is derived from these mathematical reproductions.

C.3 Correction and refusal inventory

The package preserves the infeasible coarse-grid polynomial, its continuous and exact finite counterexamples, eight numerical refinement iterations, finite-cover incumbents, failed and corrected all-length code, the independent proof receipts and malformed-input controls. Original archived outputs are not overwritten by the public reproduction command.

The wrapper refuses a stronger requested bound, a two-dimensional request, singleton classes, arbitrary selected progressions, non-exact coverage, malformed rational input, altered witness bytes, changed checker source, missing inputs, optimized Python and reuse of an existing output directory. An admitted request establishes only the theorem's conditional conclusion. It never asserts that a covering system with the supplied extra properties exists.

C.4 What "independent" means here

Parallel workstreams share a model family and one coordinated study. They are not independent human laboratories. Separation of code, derivations, inputs and review roles reduces some correlated-error risks; it does not eliminate them. The fresh conventional workstream began without access to the selected witness. The source audit checked original definitions and a stronger

prior bound, preventing an exaggerated comparison. External qualified review remains necessary before treating this research-review edition as a settled canonical reference.

Appendix D. Economic assumptions and reconciliation

All money is USD. Hours and rates below are analyst resource estimates, not observed labor or market quotations. The fixed activities sum to 110 hours; marginal activities sum to 10.5 hours per beneficiary.

Fixed activity	Hours	Assumed hourly rate
Discovery, including failed searches	24	$110
Certificate construction and repair	18	$110
Independent mathematical review	20	$140
Publication and reproduction	12	$90
Correction and failed-review allowance	12	$110
Oversight, rights and commitments	6	$120
Acquisition and coordination	10	$100
One-year shared maintenance	8	$100

Per-beneficiary activity	Hours	Assumed hourly rate
Integration	4	$110
Acceptance and independent reproduction	2	$140
Onboarding	1	$90
Support	2	$90
Correction and failure allowance	1.5	$110

Add 20% labor overhead; fixed USD300 research/verification compute and USD60 storage/distribution; and USD40 per beneficiary. General administration, tools, insurance and unallocated support are in overhead and are not charged again. Processing is 3% of fees. Discovery and construction, including compute, contribute USD5,844 once to the central project cost. All expected maintenance and correction are included before profit is computed.

Central gross end cash before financing equals initial USD5,000 plus full-project profit USD2,408. USD1,200 is encumbered by the additional warranty reserve. That reserve does not remove the remaining service duty. The six-stage ledger separately reports gross cash, restricted refundable deposits, earned revenue, paid or prepaid costs, modeled cost to complete, support prepayment and spendable cash. Infeasible unfinanced balances are labeled as projections that require stopping, not as executed spending.

At the assumed USD6,000 fee, an additional paying sponsor contributes USD4,394 before fixed costs, provided requirements and marginal workload remain unchanged. At four sponsors, a USD1,000 increase in the shared cost reduces project profit by USD1,000; a USD1,000 fee decrease per sponsor reduces profit by USD3,880 after the 3% fee. These sensitivities show why a modest positive central margin is not robust evidence of a business.

The executable model emits complete scenario, activity, cash, participation/fee, discount and arrangement tables. Its own audit executes 153 exact accounting assertions. A separately written

implementation reconstructs all three scenarios and their cash ledgers without importing the model. No scenario includes future token gains, speculative enterprise valuation or profits copied from predecessor studies.

Appendix E. Reproduction, evidence and publication status

E.1 Minimal exact reproduction

Python 3.11–3.13 with its standard library suffices for the fixed mathematical certificate. From the package root, run:

```
python final_certificate.py \
  --request study/final_request.json \
  --output local_reproduction
```

The output directory must be new. The command copies the frozen checkers into a temporary work area, recomputes both line-arrangement verifiers, their exact maximum and the tail bound, checks complete obligation coverage, and writes a new acceptance receipt. It does not rely on stored success flags alone. `python reproduce.py` additionally runs accounting reconciliation and package checks. Native ecosystem execution is optional because its unchanged dependency closure requires the documented packages.

Re-running floating-point discovery requires NumPy 2.3.2 and SciPy 1.17.0 in the recorded environment. Optimizer time limits can change incumbents across machines. The reproducible theorem is the exact rational certificate and its verification, not deterministic rediscovery by a language model. Seeds, search limits, library versions and original outputs are preserved.

E.2 Claim-to-evidence discipline

The evidence map labels claims as **executed**, **measured**, **independently checked**, **modeled**, **inferred** or **proposed**. Mathematical validity rests on the explicit proof and computational obligations. Wall-time values are local measurements. Priority is a bounded inference from inspected primary sources. Commercial fees and costs are modeled. Future services and broader infrastructure are proposed. No simulated frontier-model capability is reported.

Important files include `certificate/weight.json`, `final_certificate.py`, both line-arrangement checkers, `study/final_confirmation_protocol.json`, the earlier polynomial sources and frozen protocol, fresh confirmation receipts, the native integration manifest, `economics/results.json`, the separate cash replay, `evidence/claim_to_evidence.csv` and `evidence/prior_art.csv`. The earlier polynomial witness SHA-256 begins `c92bc1977bbb6804`; the final witness and all source hashes are recorded separately in the final freeze and package manifest. Full copyrighted research working copies are excluded; citations and authored audits remain.

E.3 Review edition and canonical release

This is a pinned research-review edition with a complete downloadable package and a browser-readable counterpart generated from the same manuscript. It is distinct from a merged canonical release, production host qualification, external peer review or customer acceptance. Repository publication must preserve existing originals and use a new edition ID. Canonical catalog promotion follows the existing paper importer and normal review/check process; executable additions retain the applicable source acceptance gates.

The repository's main source was initially inspected at `def7cdfc`; a later refresh found `3947e4ec`, the merge of the existing 1.46.1 test-edition repair. This study uses a new directory and does not take over that release owner or rewrite prior studies. Public review URLs and immutable commit identity are recorded in the separate publication receipt after verification. The LinkedIn text is an approval draft and has not been posted.

AI assistance and review disclosure. The requested author line is retained. GPT-6 Astra and parallel AI workstreams assisted source review, mathematics, code, auditing and writing. Internal cross-checks are described explicitly; no human peer review, model training, live ENS identity, paid sponsor commitment or complete billing evidence is implied.

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